MODULE 2 · STATISTICS & PROBABILITY · TOPIC 2.5

Binomial & Poisson

Day 25 — the distributions for COUNTING things



Boston University

2	6	1
Distributions	Worked examples	Question that picks between them

Prepared & presented by **Srinivasa Sai Chava**

THE SHIFT

Everything so far measured. Today we count.

Days 23 – 24 · MEASURING

- How tall is this person? 171.3 cm
- What is this salary? 42,500
- How long did delivery take? 18.4 min

The answer can be ANY number, including fractions.
These are CONTINUOUS.

Today · COUNTING

- How many of 20 items are defective? 3
- How many clicked the ad? 47
- How many calls arrived this hour? 5

The answer is always a WHOLE number: 0, 1, 2, 3...
These are DISCRETE.

Why this needs different tools

You cannot have 2.7 defective items. So instead of a smooth curve where we ask “what fraction falls between 85 and 115?”, we get separate BARS and ask “what is the probability of exactly 3?” — a question that was meaningless yesterday, because the chance of a height being exactly 171.30000… cm is zero.

Two distributions cover almost all counting problems. Today you learn both, and the one question that tells you which to use.

WARM-UP · 5 MINUTES

Four questions from Day 24

1	**What are the three bands?**	→ 68% · 95% · 99.7% within 1, 2, 3 sd.
2	**Where does 1.96 come from?**	→ ppf(0.975) — it leaves 2.5% in each tail.
3	**Two exams, which mark is better?**	→ Compare z-scores, not raw marks.
4	**The 2σ check passed. Is it normal?**	→ Not necessarily. One pass proves nothing.

Hold on to the normal curve — by the end of today you will see BOTH of today's distributions turn into it. That is not a coincidence, and Day 27 explains why.

SESSION PLAN

How today's two hours are structured

1. **What makes it binomial** — 20 min
   Four conditions, with examples that fail

2. **Building the binomial** — 25 min
   From 8 coin flips to the formula

3. **Using it** — 20 min
   pmf, cdf, mean and spread

4. **Poisson** — 25 min
   When n is huge and p is tiny

5. **Choosing between them** — 10 min
   One question decides it

6. **Practice notebook** — 30 min
   You code, I walk around

Section 2 builds the formula from scratch. If you have ever been handed it and told to memorise it, today is the session that explains where every piece comes from.

1

What Makes It Binomial

Four conditions · 20 minutes

- The four conditions
- Examples that qualify
- Examples that do NOT

A binomial needs all four of these

1	**A FIXED number of trials**	You decide n before you start. 20 items, 10 flips, 500 visitors.
2	**Each trial has TWO outcomes**	Defective or not. Clicked or not. We call them success and failure.
3	**The SAME probability every time**	p does not drift. Every item has the same 5% chance.
4	**Trials are INDEPENDENT**	One result tells you nothing about the next.

Miss any one of the four and the binomial formula gives you a wrong answer — confidently, with no error message.

Test the four conditions on real situations

✓ YES, binomial

Flip a coin 10 times, count heads

n=10 fixed · H or T · p=0.5 always · flips independent

Inspect 20 items from a large batch

n=20 · defective or not · p=5% each · batch big enough that removing one barely changes p

500 visitors see an ad, count clicks

n=500 · click or not · same p · visitors do not influence each other

× NO, not binomial

Flip until you get a head

n is NOT fixed — you stop when it happens. Condition 1 fails.

Draw 5 cards from one deck, count aces

p changes after each draw: 4/52, then 3/51… Condition 3 fails.

Did 3 family members catch flu?

Not independent — they infect each other. Condition 4 fails.

Notice each failure breaks a DIFFERENT condition. That is how to diagnose a situation: walk the four in order and find the first one that fails.

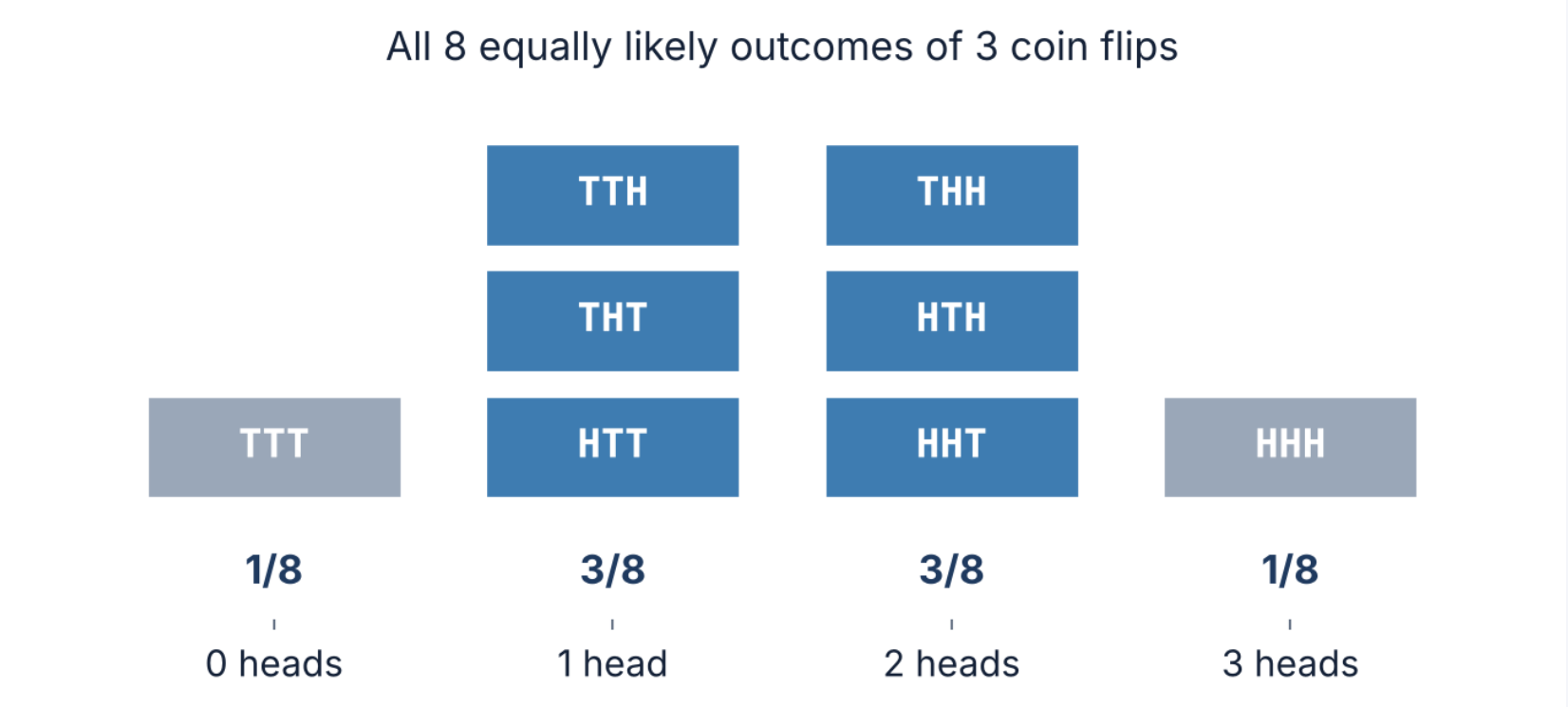
2

Building the Binomial

From 8 coin flips to the formula · 25 minutes

- List every outcome
- Spot the pattern
- The three pieces
- What n and p do

Start small: flip a coin three times



Count the ways

There is only ONE way to get zero heads: TTT.

But THREE ways to get exactly one head — the head can be in any of the three positions.

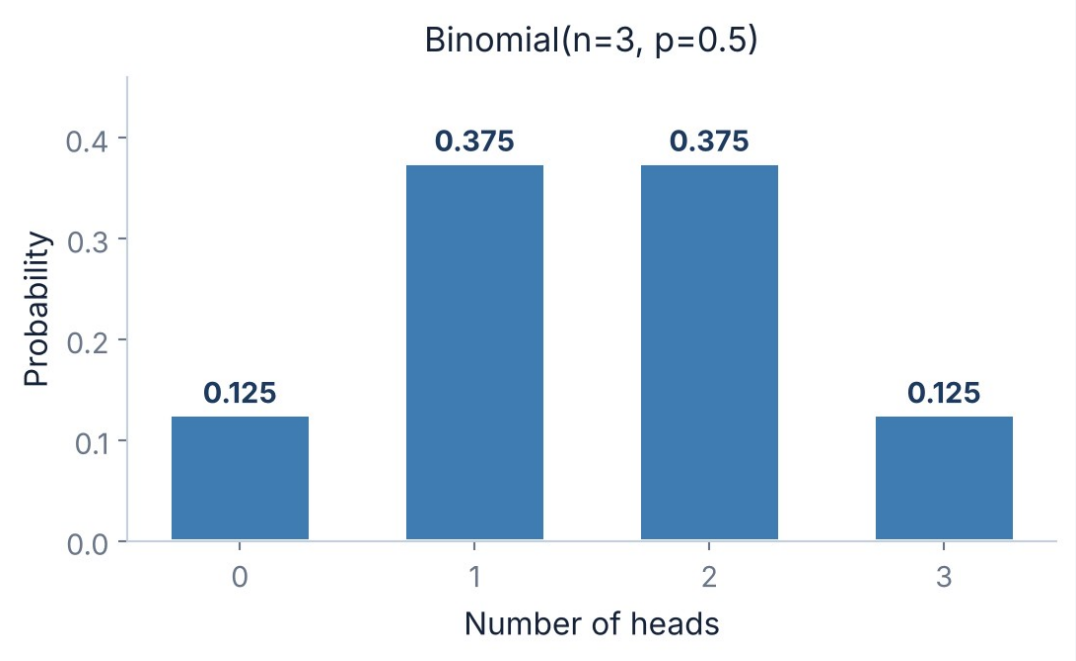
That is the whole idea.

1 3 3 1

out of 8

Every one of these 8 sequences is equally likely. So the probability of a given number of heads is just: how many sequences give that number, divided by 8.

The same eight outcomes, drawn as probabilities



1/8 = 0.125 and 3/8 = 0.375. The chart is just the counting from the previous slide, divided by 8.

Read it carefully

Each BAR is a separate possible count. There is nothing between them — you cannot get 1.5 heads, so there is no bar there.

The heights add to exactly 1.000, because one of these four things must happen.

This is what “discrete distribution” means in practice: a finite list of outcomes, each with its own probability.

```
stats.binom.pmf([0,1,2,3], n=3, p=0.5)
  ->  [0.125, 0.375, 0.375, 0.125]
```

Now the formula. Piece one: the probability of ONE path.



Multiply along the path

The trials are independent, so the probability of this exact sequence is the product of the individual probabilities:

p × (1−p) × p × p × (1−p)

which tidies up to $p^3 (1-p)^2$

And here is the key observation

Any OTHER path with 3 successes and 2 failures gives exactly the same product — multiplication does not care about order.

So every such path has probability $p^3(1-p)^2$. We just need to know how many there are.

Piece two: how many paths are there?

“Choose k positions out of n”

With 3 flips and 1 head, the head can sit in position 1, 2 or 3 — three ways. That is exactly what C(3,1) = 3 means.

```
from math import comb
comb(3, 0), comb(3, 1), comb(3, 2), comb(3, 3)
   ->   1,  3,  3,  1      ← our counts!
```

Written out

$$C(n, k) = n! / (k! \times (n-k)!)$$

You will see it called “n choose k”, and written as nCk or with a big bracket. They all mean the same thing: the number of different ways to pick k items from n when the order does not matter.

Put the two pieces together and you have the binomial formula

$$P(k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

- $C(n, k)$ — how many paths
- p^k — k successes
- $(1-p)^{(n-k)}$ — the rest are failures

Work one through by hand: 7 heads in 10 flips

$n = 10, \quad k = 7, \quad p = 0.5$

$\binom{10}{7}$	$= 120$	there are 120 different orders with 7 heads
0.5^7	$= 0.0078125$	the 7 heads
0.5^3	$= 0.125$	the 3 tails

$$120 \times 0.0078125 \times 0.125 = 0.1172$$

Check it in code

```
stats.binom.pmf(7, 10, 0.5)

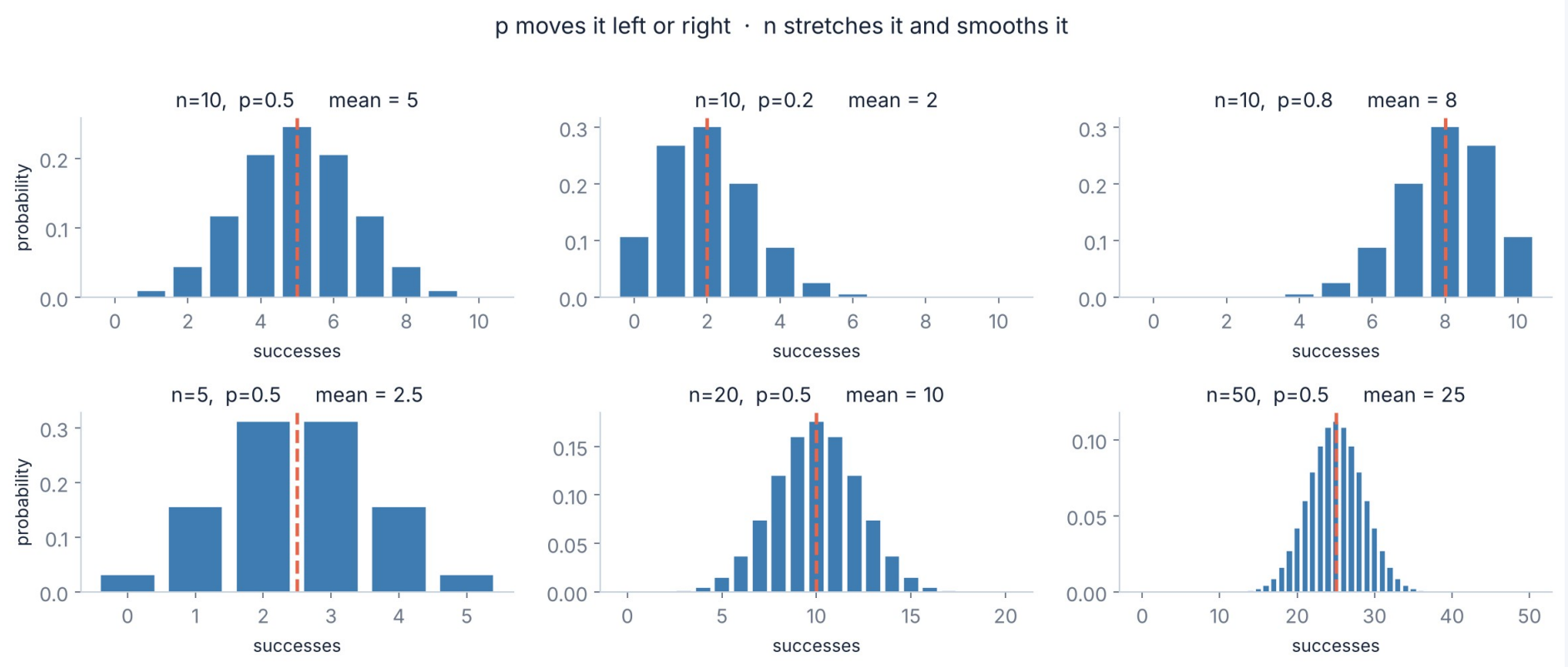
  ->  0.1171875
```

About an 11.7% chance of getting exactly 7 heads in 10 flips.

You will always use the library in practice. Doing it once by hand is so that the library is not a black box.

A useful sanity check: 7 out of 10 is above the expected 5, so the probability should be smallish but not tiny. 11.7% feels right. If you had got 0.9 or 0.001 you would know something was wrong.

What the two settings actually do to the shape



Top row: p moves the peak

Low p pushes it left, high p pushes it right. The peak sits at n×p.

Bottom row: n widens and smooths

Look at n=50: that is starting to look like a bell curve. Remember that.

3

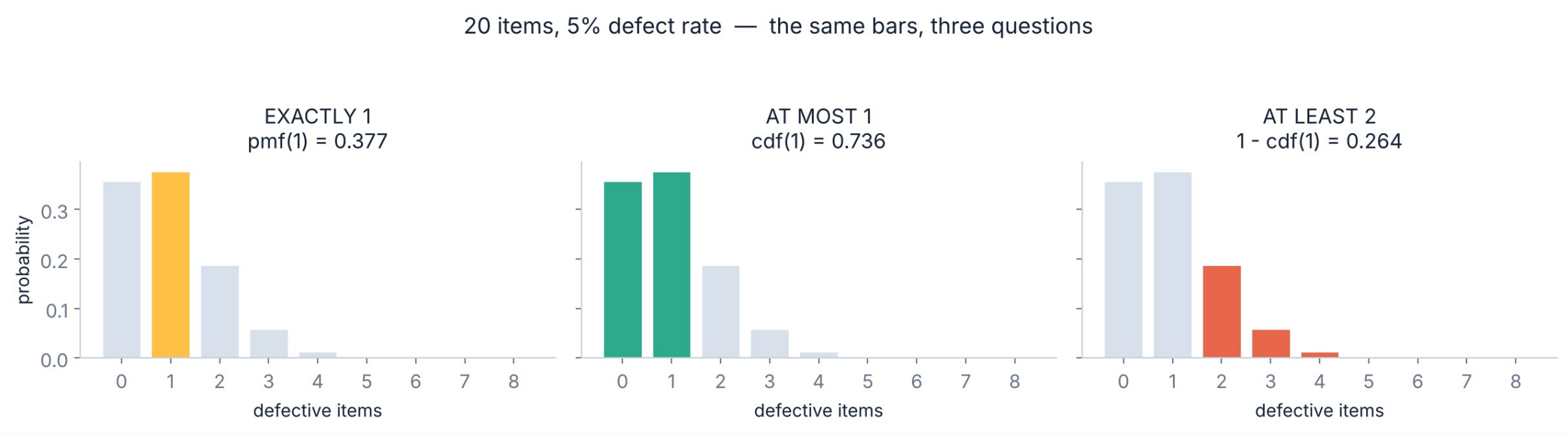
Using the Binomial

Exactly, at most, at least · 20 minutes

- pmf vs cdf
- A worked quality-control problem
- Mean and spread

Slides by Srinivasa Sai Chava

Three different questions, the same set of bars



The question	In words	Code	Answer
EXACTLY 1 defective	just that one bar	`binom.pmf(1, 20, .05)`	0.377
AT MOST 1	add up bars 0 and 1	`binom.cdf(1, 20, .05)`	0.736
AT LEAST 2	everything else	`1 - binom.cdf(1, 20, .05)`	0.264

⚠️ The mistake almost everyone makes once

“At least 2” is NOT 1 − cdf(2)

```
1 - binom.cdf(2, 20, .05)   ->  0.0755
1 - binom.cdf(1, 20, .05)   ->  0.2642
#  which one is 'at least 2'?
```

cdf(2) already INCLUDES the bar at 2. Subtracting it removes the very outcome you wanted to keep.

The rule that always works

$$P(\text{at least } k) = 1 - \text{cdf}(k - 1)$$

Say it out loud: “at least 2 means NOT 0 and NOT 1”. So subtract everything up to and including 1.

When in doubt, draw the bars and shade the ones you want. It takes ten seconds and removes all doubt.

Why this matters more than it looks

0.0755 and 0.2642 are both plausible-looking probabilities. Nothing errors, nothing warns you, and the wrong one is off by a factor of three and a half. This is the discrete-distribution equivalent of Day 16's loc-versus-iloc: an off-by-one that produces a confident, wrong number.

What to expect, and how much it varies

Two formulas worth knowing

$$\text{mean} = n \times p$$

$$\text{sd} = \sqrt{n \times p \times (1-p)}$$

The mean is obvious once you say it aloud: 20 items with a 5% defect rate should give about 1 defect.

Our quality-control example

```
n = 20,  p = 0.05

mean = 20 x 0.05          =  1.00
sd   = sqrt(20x.05x.95) =  0.97
```

So: expect about 1 defect, give or take 1.

Use these for a sanity check before you compute anything

If a batch of 20 comes back with 8 defects, you do not need a p-value to be suspicious: that is 7 above the expected 1, which is about seven standard deviations out. Day 24's ruler, applied to counts.

And notice the spread is largest when p = 0.5 and shrinks as p approaches 0 or 1 — which makes sense, because an event that almost never happens, or almost always happens, is predictable.

4

Poisson

When n is huge and p is tiny · 25 minutes

- The problem binomial cannot solve
- The rate idea
- A worked example
- The shape

A question the binomial cannot answer

“How many calls will arrive this hour?”

Try to force it into a binomial and the first question is: what is n?

How many trials are there? Every second is a chance for a call — so 3,600? Every millisecond — so 3,600,000? There is no natural answer.

And what is p? Whatever makes the numbers work out, which is not a definition.

Other questions with the same problem

- Typos per page of a book
- Accidents at a junction per month
- Customers entering a shop per 10 minutes
- Server errors per day
- Rainfall events per season

What they all have in common

Events happen at some average RATE over a continuous interval of time or space. There are no countable trials — there is just an interval, and events landing in it. You know the average; you want the probability of a particular count.

That is exactly what the Poisson distribution is for.

One number is all you need: the average rate

λ (lambda) = the average count per interval

If the call centre averages 3 calls an hour, then λ = 3.

That single number fixes the whole distribution — there is no second parameter to find. Compare the binomial, which needs both n and p.

⚠️ **λ must match your interval. 3 per hour means λ=3 for an hour, λ=1.5 for 30 minutes, λ=72 for a day.**

The formula

$$P(k) = e^{-\lambda} \times \lambda^{k} / k!$$

You will use scipy rather than this. But notice what is NOT in it: no n, no p. Only λ and the count you are asking about.

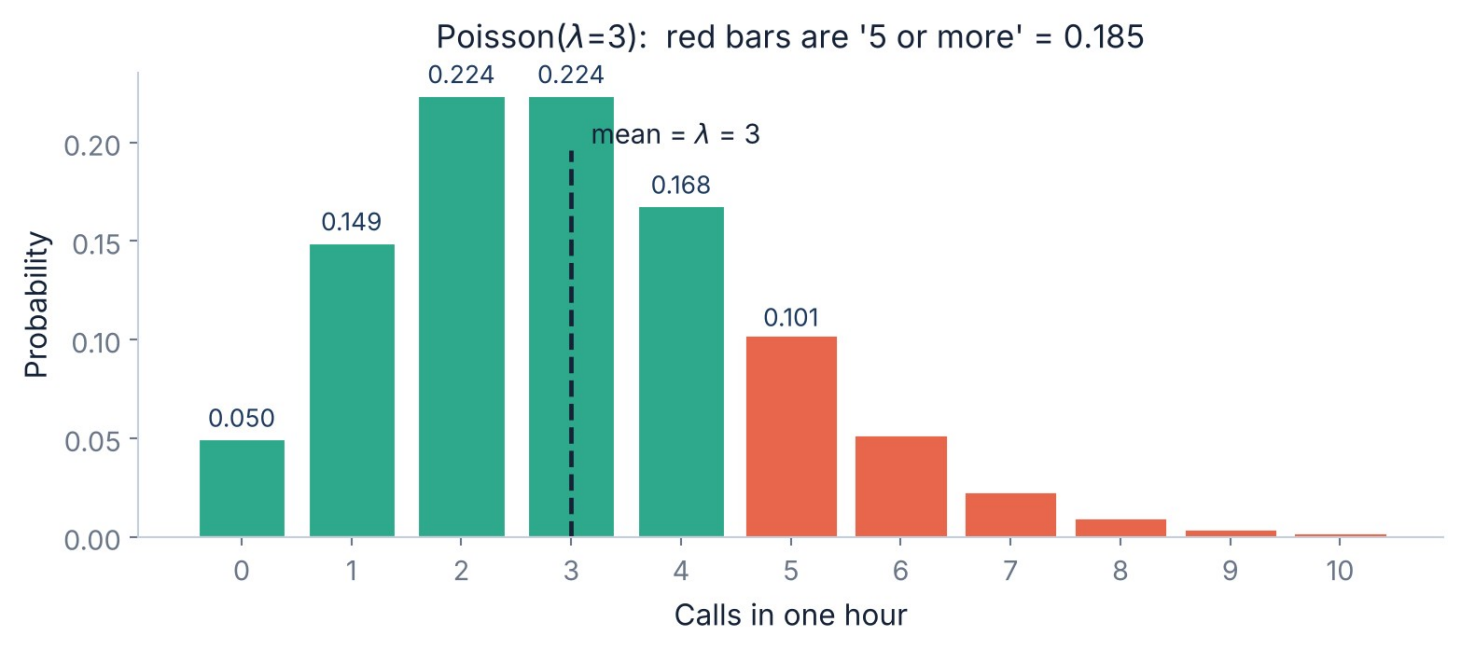
```
stats.poisson.pmf(k, mu=3)
```

The three Poisson conditions

- **Events are independent**
  one call does not cause another
- **The rate is constant**
  3 an hour all through the interval
- **Two cannot land at the same instant**
  events are separable

The constant-rate condition fails more often than people expect: a call centre at 9am and at 3am do not share a λ.

Worked through: a call centre averaging 3 an hour



Reading the chart

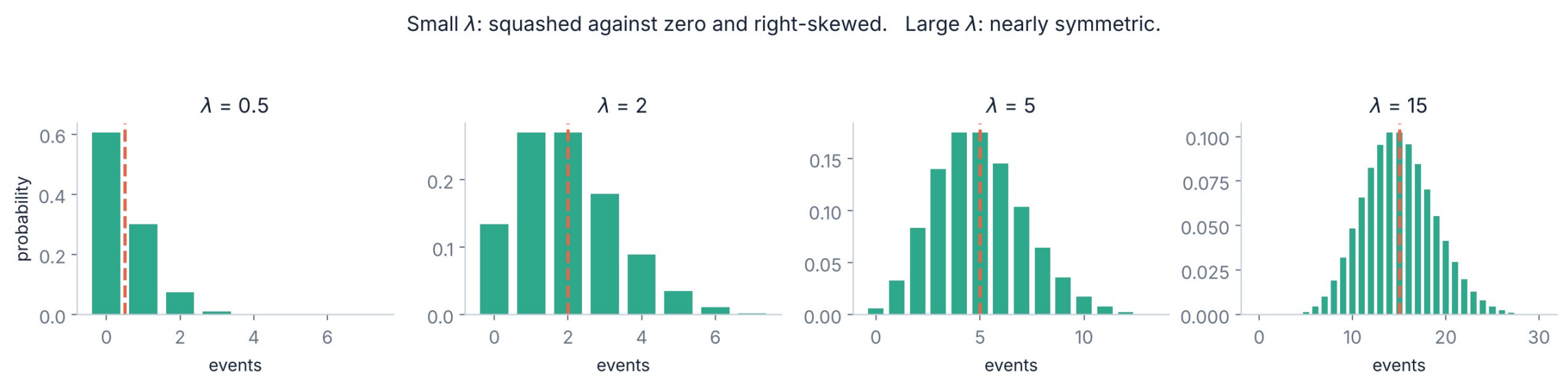
Question	Answer
no calls at all	0.050
exactly 3	0.224
at most 2	0.423
5 or more	0.185

Note 2 and 3 are equally likely — that happens whenever λ is a whole number.

```
poisson.pmf(3, 3) = 0.224
```

The staffing question this answers: 18.5% of hours will bring 5 or more calls. If you staff for the average of 3, you will be overwhelmed roughly one hour in five — which is exactly the kind of decision a distribution is for, and an average alone cannot tell you.

How λ changes the shape



Small λ: squashed against zero

Counts cannot go below zero, so with λ = 0.5 the distribution piles up at 0 and 1 and has a long right tail. Strongly skewed.

Large λ: almost symmetric

At λ = 15 the floor at zero is far away and the shape is nearly a bell curve. We will make that precise before the end of the session.

The Poisson's strangest property

$$\text{mean} = \text{variance} = \lambda$$

$$\text{so} \quad \text{sd} = \sqrt{\lambda}$$

One number fixes both the centre AND the spread. No other common distribution does this.

Why this is genuinely useful

It gives you a free diagnostic. Take any count data, compute the mean and the variance, and compare them.

If the variance is much BIGGER than the mean, the data is “overdispersed” — something is clustering the events, and Poisson is the wrong model.

The check, in one line

```
ratio = counts.var() / counts.mean()
#  near 1  -> Poisson is plausible
#  well above 1 -> events are clustering
```

Example: website visits per minute usually fail this badly. Visits arrive in bursts when a link is shared, so the variance dwarfs the mean. Earthquakes fail it too — aftershocks cluster. Both violate the independence condition, and the mean-variance ratio is how you catch it without any formal test.

5

Choosing Between Them

And where both lead · 10 minutes

- One question decides
- Binomial → Poisson
- Both → Normal

One question tells you which to use

“Can I count the number of trials?”

Everything else follows from the answer.

YES, there is a clear n → BINOMIAL

- 20 items inspected — n = 20
- 10 coin flips — n = 10
- 500 visitors shown an ad — n = 500
- A class of 60 sitting an exam — n = 60

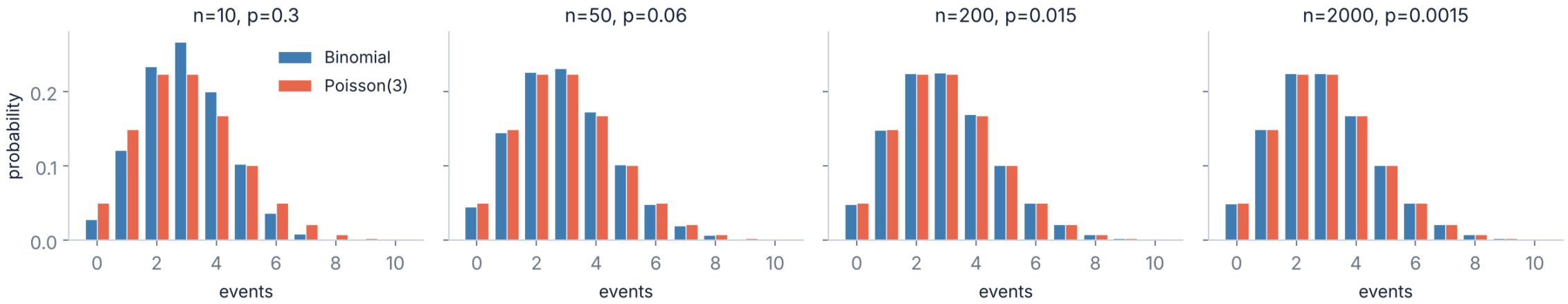
NO, just a rate → POISSON

- Calls per hour — no n
- Typos per page — no n
- Accidents per month — no n
- Emails per day — no n

A useful tell: if the question says “out of”, it is binomial. If it says “per”, it is Poisson.

They are not really two different things

Same mean (n×p = 3) throughout. As n grows and p shrinks, the two become identical.



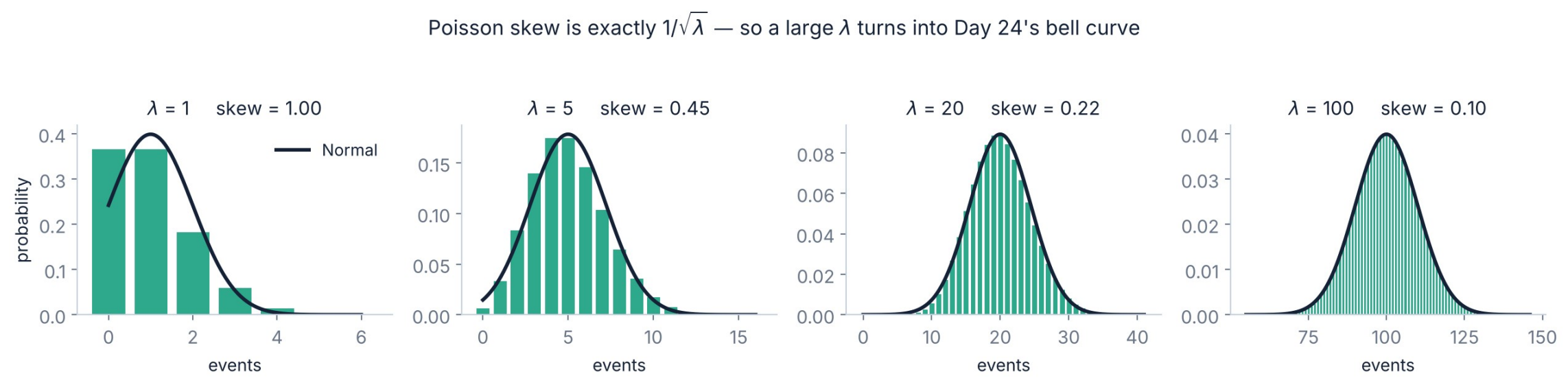
Watch the orange and blue bars merge

At n=10 they differ visibly. By n=2000 they are indistinguishable. Poisson is what the binomial BECOMES when n grows and p shrinks with n×p held fixed.

Which is why the rule of thumb exists

If n > 50 and p < 0.05, you may use Poisson with λ = n×p. In practice you will just use the binomial — computers do not care — but it explains why Poisson works for rare events at all.

And both of them become yesterday's bell curve



The skew of a Poisson is exactly 1 / √λ

At λ=1 the skew is 1.00 — badly skewed. At λ=100 it is 0.10 — all but symmetric, and the black normal curve sits right on top of the bars. The same thing happens to the binomial as n grows, which is what you noticed in the n=50 panel earlier.

That is twice in one session that the normal curve has turned up uninvited. Day 27 explains why it keeps happening.

The two distributions side by side

	Binomial	Poisson
Question it answers	how many successes OUT OF n?	how many events PER interval?
Parameters	n and p	λ only
Possible values	0 to n	0 upward, no limit
Mean	n × p	λ
Variance	n × p × (1−p)	λ (same as the mean)
scipy	`stats.binom`	`stats.poisson`
Exactly k	`binom.pmf(k, n, p)`	`poisson.pmf(k, lam)`
At most k	`binom.cdf(k, n, p)`	`poisson.cdf(k, lam)`
At least k	`1 - binom.cdf(k-1, ...)`	`1 - poisson.cdf(k-1, ...)`

Both are discrete, so both use pmf for “exactly” and cdf for “at most”. Neither has a pdf — that is for continuous distributions like yesterday's normal.

RECAP

Debug corner — eight counting mistakes

You did	What goes wrong	Fix
used 1 - cdf(k) for “at least k”	off by one — excludes k itself	1 - cdf(k-1)
binomial on cards without replacement	p changes every draw	hypergeometric distribution
binomial when n is not fixed	“flip until a head” has no n	geometric distribution
λ for the wrong interval	3/hour is not 3 per 20 minutes	scale λ to your window
Poisson when events cluster	variance far exceeds the mean	check var/mean before modelling
pdf on a discrete distribution	there is no such method	pmf for counts, pdf for continuous
forgot p in binom.pmf(k, n)	TypeError, or a silent default	always pass n AND p
asked P(exactly 2.5)	counts are whole numbers	the question is malformed

Row 1 is the one that will cost you marks. Row 4 is the one that will cost you a production incident.

Putting it together — a staffing decision

```python
from scipy import stats

#  A support desk averages 3 calls an hour.
#  One agent can handle 4 calls an hour.
#  QUESTION: how often are we overwhelmed?

lam = 3

#  --- 1. the risk with one agent
overwhelmed = 1 - stats.poisson.cdf(4, lam)
print(f"P(5 or more calls) = {overwhelmed:.3f}")
#  0.185  -> about one hour in five

#  --- 2. what would two agents give us?
#  two agents handle 8 an hour
print(1 - stats.poisson.cdf(8, lam))
#  0.0038 -> about one hour in 260

#  --- 3. express it as hours per month
hours = 8 * 22                      # a working month
for agents in (1, 2, 3):
    cap = 4 * agents
    risk = 1 - stats.poisson.cdf(cap, lam)
    print(f"{agents} agent(s): overwhelmed {risk*hours:5.1f} hours/month")
```

λ matches the interval

3 per hour, asked per hour

cdf(4), not cdf(5)

“5 or more” = 1 - cdf(4)

A probability, not a count

0.185 of hours, not 0.185 calls

Translate to a unit people feel

“hours per month” beats 0.185

Compare the options

the decision needs alternatives

The average alone is useless

3 < 4, so “we are fine” — wrong

The last note is the real lesson: the average load is 3 and capacity is 4, so an average-based answer says “no problem”.

OVER TO YOU

Practice session — 30 minutes

A **Predict the output** — 8 questions

pmf, cdf, at-least, mean, λ scaling

B **Spot & fix the bug** — 4 questions

Off-by-one, wrong interval, wrong model

C **Write the code** — 5 tasks

Build, compute, compare, decide

D **Challenge** — 2 tasks

A staffing decision, and a model that fails

Your notebook

`Day25_Practice_Questions.ipynb`

- Ask: “out of” or “per”?
- Draw the bars when unsure
- Raise a hand — I am walking around
- Solutions shared after class

For every answer, say what it means in words. “0.185” is not an answer; “about one hour in five” is.

TODAY IN TWELVE LINES

Recap

1. Days 23–24 measured quantities. Today we COUNT events.
2. Counts are whole numbers, so we get bars, not a curve.
3. Binomial needs four things: fixed n, two outcomes, same p, independence.
4. Its formula is C(n,k) × p^k × (1−p)^(n−k) — paths × successes × failures.
5. C(n,k) is just “how many orders give k successes”. For n=3: 1, 3, 3, 1.
6. Binomial mean = np, sd = √(np(1−p)).
7. ⚠ “At least k” is 1 − cdf(k−1). Using cdf(k) is the classic error.
8. Poisson is for a RATE with no countable n: calls per hour, typos per page.
9. It needs one number, λ — and λ must match your interval.
10. Poisson mean = variance = λ. If var ≫ mean, events are clustering.
11. The deciding question: “out of” → binomial, “per” → Poisson.
12. Binomial → Poisson as n grows; both → Normal. Day 27 says why.

Homework

Write out all 16 outcomes of 4 coin flips and check they give C(4,k) · Find three situations at your own workplace or studies, one binomial and one Poisson, and say which condition decided it · Redo the staffing build with λ = 8

Next class — Topic 2.6

The uniform distribution — when every outcome is equally likely. The simplest distribution of all, and the one every random number generator is built on.